

Dilated Cardiomyopathy 2B (CMD2B / GATAD1-related DCM)

Disease knowledge-base research report MONDO:0013848 · OMIM 614672 · Gene: GATAD1 (7q21.2) Date: 2026-09-25

Evidence-base caveat. CMD2B is an ultra-rare Mendelian cardiomyopathy. The human clinical description rests essentially on **a single consanguineous kindred** (Theis et al. 2011,

P 21965549).
Almost all mechanistic detail comes from **in vitro biochemistry** (
P 38605029
)
P 26841866)
and **model organisms** (zebrafish
;
P 28955713
mouse

P 39641830
)
P 39626862).

Where a statement is generic to dilated cardiomyopathy (DCM) as a class rather than specific to CMD2B, this is flagged. Source type is labeled throughout as [human clinical], [in vitro], [model organism], or [computational/inferred].

1. Disease Information

Overview. Dilated Cardiomyopathy 2B (CMD2B) is a rare **autosomal recessive** form of familial dilated cardiomyopathy caused by biallelic mutation of **GATAD1** (GATA zinc-finger domain-containing protein 1). Like other DCMs it is defined by **left-ventricular (LV) dilation and systolic dysfunction** (reduced ejection fraction) not explained by abnormal loading conditions or coronary disease, presenting clinically as heart failure and predisposing to arrhythmia. CMD2B is distinguished from the far more common autosomal-dominant DCMs by its recessive transmission with clinically silent heterozygous carriers [human clinical, [P 21965549](#)].

[P 21965549](#)

Key identifiers. | Resource | ID | |---|---| | Mondo | **MONDO:0013848** (dilated cardiomyopathy 2B) | | OMIM (phenotype) | **614672** — CARDIOMYOPATHY, DILATED, 2B; CMD2B | | OMIM (gene) | **614518** — GATAD1 | | HGNC | **HGNC:29941** (GATAD1) | | NCBI Gene | **57798** | | Ensembl | ENSG00000157014 | | UniProt | Q8WUU5 (GATD1_HUMAN) | | RefSeq | NM_021167.5 / NP_066990.3 (GATAD1, 249 aa); LRG_746 | | Founder variant | NM_021167.5:c.304T>C (p.Ser102Pro); ClinVar VCV31656; rs387907188 | | ICD-10 | I42.0 (Dilated cardiomyopathy) | | ICD-11 | BC43.0 (Dilated cardiomyopathy) | | MeSH | D002311 (Cardiomyopathy, Dilated) | | Orphanet | within ORPHA:154 (Familial isolated dilated cardiomyopathy) — no distinct subtype code |

Synonyms / alternative names. CMD2B; Cardiomyopathy, dilated, 2B; GATAD1-related/GATAD1-associated dilated cardiomyopathy; autosomal recessive dilated cardiomyopathy (GATAD1 type).

Data source type. Disease-level aggregated resources (OMIM/Mondo) plus **individual-patient** data from one published family; no EHR/registry-scale cohort exists for this specific subtype.

2. Etiology

Primary causal factor — genetic. CMD2B is a **monogenic, biallelic** disorder. Disease requires two loss-/alteration-of-function alleles of **GATAD1**. In the index family a homozygous missense mutation altering the conserved **serine-102** residue (reported as **p.Ser102Pro**) segregated with

recessive DCM; it was absent from HapMap, 1000 Genomes and 474 ethnically matched controls [human clinical,].

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"altered a conserved residue of GATAD1 ... Thirteen relatives were heterozygous mutation carriers with no evidence of myocardial disease, even at advanced ages."

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Genetic risk factors. - Causal variant: homozygous **GATAD1 p.Ser102Pro** (7q21.2). Because unaffected heterozygotes exist, a single allele is insufficient (recessive). - **Consanguinity** is a major predisposing structural factor — the founding couple were first cousins, and homozygosity mapping exploited runs of homozygosity [human clinical,].

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- Modifier genes: none specifically identified for CMD2B (data unavailable).

Environmental / lifestyle risk factors. No CMD2B-specific environmental trigger is established. For DCM broadly, alcohol, cardiotoxic chemotherapy (anthracyclines), myocarditis/viral infection, peripartum state, thyroid disease and tachyarrhythmia are recognized acquired causes/"second hits"; these may act as **gene–environment modifiers** superimposed on genetic susceptibility [DCM-class;

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notes "second-hit" genetic contributions]. Not demonstrated for GATAD1 specifically.

Protective factors. No genetic or environmental protective factor is described for CMD2B (data unavailable). By inference, absence of a second pathogenic allele is "protective" (heterozygotes unaffected).

Gene–environment interactions. Mouse data suggest GATAD1 becomes functionally important under **stress**: cardiac Gatad1 loss worsens ischemia–reperfusion injury and abolishes sphingosylphosphorylcholine cardioprotection [model organism,],

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implying environmental/metabolic stressors may unmask or aggravate GATAD1-deficient myocardium — inferred, not shown in patients.

3. Phenotypes

All descriptions derive from the single reported kindred [human clinical,

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plus generic DCM knowledge.

Phenotype	Type	HPO term	Onset / severity / course	Frequency
Dilated cardiomyopathy	clinical/ imaging sign	HP:0001644	Adult-onset, progressive	Core, defining (homozygotes)
Left ventricular enlargement / dilatation	imaging sign	HP:0001640 (cardiomegaly), HP:0001712 (LV hypertrophy—N/A)	Adult; may be isolated/ milder	Present incl. an "intermediate" male
Reduced LV ejection fraction / systolic dysfunction	imaging/ functional	HP:0001635 (congestive heart failure), HP:0005162 (abnormal LV function)	Progressive	Core
Congestive heart failure (dyspnea, fatigue, edema)	symptoms	HP:0001635	Adult, progressive	Expected in overt cases
Arrhythmia / risk of sudden death	sign	HP:0011675 (arrhythmia)	Variable	DCM-class risk; not detailed for CMD2B
Abnormal cardiomyocyte nuclear morphology	pathology (biopsy)	(cellular)	—	Seen in proband myocardium

Phenotype characteristics. Age of onset: **adult-onset** (HP:0003581) with **age-dependent penetrance**; the affected sisters were adults, and a brother had milder "idiopathic LV enlargement," illustrating **variable expressivity** (HP:0003828). Severity ranges from isolated LV dilatation to overt DCM/heart failure. Course: **chronic, progressive**.

Quality-of-life impact. No CMD2B-specific QoL data. For symptomatic DCM/heart failure generally, health-related QoL is substantially reduced (dyspnea, exertional limitation, hospitalizations); disease-specific instruments include the Kansas City Cardiomyopathy Questionnaire and Minnesota Living with Heart Failure Questionnaire (data unavailable for this subtype).

4. Genetic / Molecular Information

Causal gene. **GATAD1** — GATA zinc-finger domain-containing protein 1; 7q21.2; HGNC:29941; NCBI Gene 57798; OMIM 614518; UniProt Q8WUU5. Encodes a nuclear protein containing a **GATA-type zinc finger** that functions as a **reader of H3K4me3** and a subunit of a histone-modifying chromatin complex [in vitro, [P 26841866](#) 39641830].

Pathogenic variant (founder allele of the index family). - **HGVS:** **GATAD1 NM_021167.5:c.304T>C**, protein **p.Ser102Pro** (LRG_746; UniProt Q8WUU5:p.Ser102Pro); genomic **chr7:g.92078120T>C (GRCh38)** [database-verified, ClinVar]. - **Identifiers:** **ClinVar Variation ID 31656; dbSNP rs387907188.** - **Type/class:** single-nucleotide **missense; germline;** homozygous in affected individuals [human clinical, [P 21965549](#)].

- **Classification:** **Pathogenic** in ClinVar for "Dilated cardiomyopathy 2B" (review status: *no assertion criteria provided* — 1-star). Ser102 is N-terminal to the GATA-type zinc finger, within the intrinsically disordered region bearing the 14-3-3 phospho-motif. In-silico: **CADD 21.8, PolyPhen-2 probably damaging;** SIFT tolerated (mildly discordant). - **Allele frequency:** **absent from gnomAD** (genomes and exomes) and absent from HapMap, 1000 Genomes and 474 matched controls [PMID 21965549] — private/ ultra-rare. - **Broader GATAD1 landscape:** most GATAD1 ClinVar entries are **VUS or likely benign;** only rare truncating variants are (likely) pathogenic. GATAD1 is frequently reported as **VUS** on cardiomyopathy panels

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— caution for novel variants. - **Functional consequence:** the mutation destroys a **phosphoserine 14-3-3 docking site**, altering GATAD1 nucleocytoplasmic transport and producing aberrant subcellular localization and abnormal nuclear morphology [in vitro,

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human

IHC,

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Mechanistically this is **not a simple null** (see §6): a cardiomyocyte Gatad1 knockout mouse is healthy [PMID 39641830], suggesting an aberrant-function/mislocalization effect rather than pure loss-of-function.

Modifier genes / epigenetic / chromosomal. No CMD2B-specific modifier genes, constitutional epigenetic marks, or chromosomal abnormalities reported (data unavailable). Note GATAD1 itself acts within an **epigenetic (histone-modifying)** complex — see §6.

5. Environmental Information

No environmental factor, occupational exposure, toxin, or infectious agent is implicated in the causation of **CMD2B specifically** (it is a monogenic disease; data unavailable). Generic DCM environmental contributors (alcohol, anthracyclines, viral myocarditis, peripartum state) may act as aggravating **second hits** on a genetically susceptible myocardium but are unproven for GATAD1. No infectious agent applies.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Homozygous GATAD1 p.Ser102Pro** substitutes proline for a serine phosphorylation site → **abolishes a phospho-Ser 14-3-3 docking motif** [in vitro,

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2. Loss of 14-3-3 binding → **dysregulated masking of a nuclear localization signal** → **altered nucleocytoplasmic transport / mislocalization** of GATAD1 [in vitro,

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observed as **aberrant subcellular localization and abnormal nuclear morphology** in patient LV myocytes [human clinical,

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3. Mislocalized/altered GATAD1 → **disrupted incorporation/function within the EMSY–KDM5A(H3K4me3 demethylase)–SIN3B/HDAC histone-modifying complex** [in vitro,

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] → **altered H3K4me3-linked transcriptional regulation** at target promoters [inferred].
4. **Branch A (transcriptional/chromatin):** dysregulated cardiac gene expression programs → impaired maintenance of normal cardiomyocyte structure/function → **contractile dysfunction** [inferred from

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5. **Branch B (metabolic):** as a transcription factor GATAD1 normally represses fatty-acid-oxidation genes (**Acaa2**, **Acadm**) and promotes glucose oxidation; its dysfunction **shifts myocardial substrate metabolism and lowers stress tolerance** (larger infarcts / worse function under ischemia–reperfusion) [model organism,

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6. Converging branches → **cardiomyocyte dysfunction** → **LV chamber dilation and systolic dysfunction (reduced EF)** [human clinical,

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7. → **congestive heart failure**, and (DCM-class) predisposition to **ventricular arrhythmia / sudden cardiac death** → end-stage disease potentially requiring transplantation.

Steps 1–2 are experimentally demonstrated; step 3 (complex membership) is demonstrated biochemically but not in heart tissue; steps 4–7 are **inferred** from correlative human pathology and model-organism data. A key unresolved point: cardiomyocyte-specific **Gatad1 deletion does not cause disease in mouse**

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so the pathogenic mechanism likely involves an **aberrant/ dominant-negative-like** effect of the mislocalized mutant and/or non-cell- autonomous or developmental requirements — not simple loss-of-function.

Detail by category

- **Molecular pathways:** chromatin/epigenetic regulation via H3K4me3 reading/demethylation (KDM5A) and Sin3/HDAC deacetylation; 14-3-3 signaling. GO: **GO:0006325** (chromatin organization), **GO:0034968** (histone lysine methylation), **GO:0032452** (histone demethylase activity), **GO:0000122** (negative regulation of transcription).
- **Cellular processes:** transcriptional regulation, nucleocytoplasmic transport (**GO:0006913**), cardiac muscle contraction; metabolic gene regulation.
- **Protein dysfunction:** loss of a regulatory phosphosite → altered protein–protein interaction (14-3-3) and mislocalization; GATAD1 is intrinsically disordered in the relevant region [in vitro,

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- **Metabolic changes:** dysregulated balance of **fatty-acid oxidation (GO:0019395)** vs **glucose oxidation** [model organism,

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- **Immune involvement:** none established (non-inflammatory genetic cardiomyopathy).

- **Tissue-damage mechanisms:** DCM-class remodeling — cardiomyocyte dysfunction, interstitial/replacement fibrosis, chamber dilation; increased ischemic vulnerability [inferred/model].
- **Epigenetic changes:** GATAD1 is itself part of the epigenetic machinery (H3K4me3 reader). Disease-associated global methylation changes: data unavailable.
- **Cell types:** cardiac muscle cell / cardiomyocyte — **CL:0000746** (ventricular cardiomyocyte **CL:0002131**).

7. Anatomical Structures Affected

- **Organ (primary):** heart — **UBERON:0000948**; specifically **left ventricular myocardium / heart left ventricle** — **UBERON:0002084** (myocardium **UBERON:0002349**).
- **Secondary/systemic:** cardiovascular system (**UBERON:0004535**); congestive heart failure secondarily affects lungs (pulmonary congestion), kidneys, liver (congestion). Body system: **cardiovascular**.
- **Tissue/cell:** striated **cardiac muscle** tissue; **cardiomyocytes (CL:0000746 / ventricular CL:0002131)** are the primary affected cell population.
- **Subcellular: nucleus (GO:0005634)** — GATAD1 is nuclear; the defect centers on nuclear import/nuclear morphology. Chromatin (**GO:0000785**). A metabolic arm implicates **mitochondrial** fatty-acid oxidation machinery indirectly (**GO:0005739**) [inferred].
- **Localization / laterality:** predominantly **left ventricular**; DCM is typically **global/bilateral** ventricular involvement rather than focal.

8. Temporal Development

- **Onset: adult-onset** (**HP:0003581**), **insidious/chronic**; **age-dependent penetrance** (an explicit rationale for building an adult zebrafish model,

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No neonatal/pediatric CMD2B cases reported.

- **Progression:** chronic and **progressive**; ranges from isolated LV enlargement (intermediate phenotype) to overt DCM with systolic dysfunction and heart failure [human clinical,

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DCM-class staging: early (asymptomatic LV dilation) → symptomatic HF → advanced/end-stage.

- **Course pattern:** progressive, lifelong (chronic). No spontaneous remission documented for CMD2B; DCM broadly can show reverse remodeling with guideline-directed therapy.
- **Critical periods:** adulthood is the window of clinical expression; the pre-symptomatic phase is the opportunity for surveillance/early therapy in at-risk homozygotes.

9. Inheritance and Population

- **Inheritance: autosomal recessive** (a defining, unusual feature — most DCM is autosomal dominant). Heterozygous carriers are unaffected even at advanced age [human clinical,

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- **Penetrance:** appears high in homozygotes but **age-dependent; carriers non-penetrant**.
- **Expressivity: variable** (isolated LV enlargement → overt DCM within one family).
- **Consanguinity:** central — index family descended from first cousins; homozygosity mapping was the discovery strategy [PMID 21965549].
- **Founder effect / geographic variant distribution / anticipation / germline mosaicism:** none reported (data unavailable). No repeat expansion (no anticipation expected).
- **Carrier frequency:** not established; the specific allele is private/ultra-rare (absent from population databases).
- **Epidemiology:** No prevalence/incidence figures exist for CMD2B specifically — it is **exceedingly rare** (essentially one published kindred). For context, **DCM overall** has an estimated prevalence on the order of **~1 in 250–500** and a monogenic cause is identifiable in **~30–40%** of familial cases

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GATAD1 accounts for only a tiny fraction. Sex ratio: DCM overall is male-predominant; both sexes affected in the CMD2B family (the two fully affected probands were female). Age distribution: adults.

10. Diagnostics

Cardiac phenotyping (DCM-class, applied to CMD2B). - **Imaging:** transthoracic echocardiography (LV dilation, reduced LVEF) — first-line; **cardiac MRI** (chamber volumes, function, late-gadolinium fibrosis) [DCM-class; stress echo relevance

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LOINC/RadLex apply. - **ECG / Holter:** arrhythmia and conduction assessment. - **Biomarkers:** NT-proBNP/BNP (heart-failure severity); troponin as adjunct (non-specific). No CMD2B-specific circulating biomarker. - **Endomyocardial biopsy:** not routine; in the proband it revealed **aberrant cardiomyocyte nuclear morphology / GATAD1 mislocalization** [PMID 21965549].

Genetic testing (definitive for subtype). - Recommended approach: **NGS cardiomyopathy multigene panel** (GATAD1 is included on modern panels,

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or **whole-exome/genome sequencing**; the original diagnosis used **homozygosity mapping + exome sequencing** in a consanguineous pedigree [PMID 21965549]. - Confirm **biallelic** GATAD1 variants (homozygous or compound heterozygous) with segregation; classify per ACMG/AMP (many GATAD1 variants are currently VUS). - CMA/karyotype/FISH/mtDNA/repeat-expansion testing: **not indicated** for this single-gene point-mutation disorder.

Clinical criteria / differential diagnosis. Diagnose DCM by LV dilation + systolic dysfunction after excluding ischemic, valvular, hypertensive, toxic (alcohol/anthracycline), infiltrative, peripartum and myocarditic causes. Differential: other genetic DCMs (TTN, LMNA, RBM20, FLNC, DES, MYH7), arrhythmogenic and hypertrophic cardiomyopathies. The recessive pattern + consanguinity + GATAD1 genotype distinguishes CMD2B.

Screening. Cascade genetic testing of relatives and **echocardiographic surveillance** of at-risk biallelic individuals (see §13). Family screening yields ~10% at baseline and ~10% more over 5 years in DCM generally

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11. Outcome / Prognosis

- **CMD2B-specific outcome data are essentially absent** beyond the index family (adult-onset DCM/heart failure). No survival, mortality or QoL statistics exist for this subtype.
- **By DCM class:** prognosis depends on LVEF and remodeling. In a large registry, patients presenting with **mid-range EF (40–49%)** had markedly better outcomes than reduced-EF (<40%) patients — death/transplant **9% vs 36%** and SCD/major ventricular arrhythmia **4.5% vs 15%** over ~10 years
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);
~17% of mid-range progressed to reduced EF.
- **Complications:** progressive heart failure, ventricular arrhythmia/**sudden cardiac death**, atrial fibrillation, thromboembolism, and end-stage disease requiring **heart transplantation** (genetic DCMs are enriched among transplant recipients,

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- **Recovery:** reverse remodeling possible with guideline-directed therapy in DCM broadly; not documented specifically for CMD2B.
- **Prognostic factors:** LVEF, LV size, fibrosis on MRI, arrhythmia burden, NT-proBNP (DCM-class). No validated CMD2B-specific prognostic biomarker.

12. Treatment

No gene-specific or curative therapy exists for CMD2B. Management follows **guideline-directed medical therapy (GDMT) for DCM/heart failure with reduced ejection fraction:**

- **Pharmacotherapy (NCIT terms in brackets):**
 - **ACE inhibitors / ARBs / ARNI** (sacubitril–valsartan) — afterload/neurohormonal.
 - **Beta-blockers** (carvedilol, metoprolol succinate, bisoprolol).
 - **Mineralocorticoid-receptor antagonists** (spironolactone, eplerenone).
 - **SGLT2 inhibitors** (dapagliflozin, empagliflozin).
 - **Diuretics** for congestion; anticoagulation if AF/thrombus.
- NCIT: *ACE Inhibitor, Beta-Adrenergic Blocker, Diuretic, Aldosterone Antagonist.*

- **Pharmacogenomics:** no GATAD1-specific PGx. General HF PGx (e.g., warfarin CYP2C9/VKORC1) applies only if those drugs are used.
- **Device / interventional:** **ICD** for sudden-death prevention and **CRT** for conduction delay per EF/QRS criteria; mitral intervention as indicated.
- **Advanced / surgical:** **mechanical circulatory support (LVAD)** and **heart transplantation** for end-stage disease.
- **Advanced therapeutics (gene/cell/RNA/targeted/immunotherapy):** **none approved or in trials for GATAD1-DCM** (data unavailable). Gene-directed strategies are conceptual only.
- **Supportive/rehabilitative:** cardiac rehabilitation, sodium/fluid guidance, exercise counseling, HF self-management.
- **Experimental / trials:** no CMD2B-specific registered trials identified.
- **Personalized approach:** genotype confirmation guides **family screening and reproductive counseling** rather than drug selection at present.

13. Prevention

- **Primary prevention:** not possible for the genetic cause; **preconception/ reproductive genetic counseling** in consanguineous or carrier families is the main lever. Options: carrier testing of partners, **prenatal diagnosis** or **preimplantation genetic testing (PGT-M)** for known biallelic risk.
- **Secondary prevention:** **cascade genetic testing** of relatives and **serial echocardiographic surveillance** of biallelic (homozygous/compound-het) at-risk individuals to detect subclinical LV dysfunction early; genotype-guided screening is efficient because non-carriers of the familial variant do not develop disease (

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ACMG/HFSA recommend genetic evaluation + family screening in familial DCM.
- **Tertiary prevention:** GDMT, ICD for arrhythmic death, HF-hospitalization avoidance, management of AF/thromboembolism.
- **Behavioral:** avoid alcohol excess and cardiotoxins; standard cardiovascular risk-factor control (may mitigate "second-hit" aggravation — inferred).
- **Counseling:** autosomal-recessive counseling — **25% recurrence** risk for future sibs of an affected child; offspring of an affected person are obligate carriers (affected only if partner

is also a carrier). Immunization/public-health and environmental interventions: not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** GATAD1 is evolutionarily conserved.
- Human **GATAD1** — NCBI Gene **57798**, HGNC:29941, 7q21.2 (*Homo sapiens*, Taxon **9606**).
- Mouse *Gatad1* — NCBI Gene **67210** (*Mus musculus*, NCBI Taxon **10090**; chr5 A1).
- Zebrafish *gatad1* — NCBI Gene **678617** (*Danio rerio*, NCBI Taxon **7955**).
- **Natural disease in other species:** No naturally occurring GATAD1 cardiomyopathy in companion animals or wildlife is catalogued (no OMIA entry identified; data unavailable). Note: naturally occurring **DCM** is common in dogs (e.g., Doberman Pinscher, Great Dane) but is genetically distinct (not GATAD1).
- **Comparative biology / conservation:** the H3K4me3-reader/chromatin-complex function and the Ser102/14-3-3 regulatory motif are conserved, enabling cross-species modeling; however, **phenotypic conservation is incomplete** — zebrafish reproduce DCM whereas mouse cardiomyocyte knockout does not (§15).
- **Transmission / zoonosis:** not applicable (non-communicable genetic disease).

15. Model Organisms

Zebrafish (*Danio rerio*, Taxon 7955) — supportive positive model. Yang, Shah, Olson & Xu 2016

(P 28955713)

generated an **adult zebrafish** model of the *gatad1* homologue specifically to overcome age-dependent penetrance and validate GATAD1 as a bona fide DCM gene in a higher-throughput vertebrate.

"we generated an adult zebrafish model, which is a simpler vertebrate model with higher throughput than rodents."

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- Model type: genetic vertebrate; application: validating causality, studying adult-onset cardiomyopathy mechanisms.

Mouse (*Mus musculus*, Taxon 10090) — informative negative / limitation. Pang et al. 2024

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P 39641830)

made a **cardiomyocyte-specific *Gatad1* knockout (cKO)**. It showed **normal cardiac function to 18 months, normal nuclear shape** (unlike patients), and **normal response to pressure overload (TAC)**.

"deletion of *Gatad1* in cardiomyocytes does not induce cardiomyopathy during aging or affect the response to pressure-overload stress in mice."

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- Interpretation/limitation: pure cardiomyocyte loss-of-function is insufficient in mouse → argues the human p.Ser102Pro acts via aberrant function/mislocalization or non-cardiomyocyte/developmental requirements; a **knock-in of the p.Ser102Pro allele** would be the more faithful (still-needed) model.

Additional mouse model (metabolic). A myocardium-specific *Gatad1* cKO was used to show GATAD1 controls the fatty-acid/glucose oxidation balance and modulates ischemia–reperfusion injury

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In vitro / cellular systems. Peptide-interaction proteomics and structural studies of the Ser102 phosphosite–14-3-3 interaction

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biochemical characterization of the EMSY/KDM5A/SIN3B complex

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No iPSC- cardiomyocyte CMD2B model is yet reported (opportunity/gap).

Resources: MGI (mouse *Gatad1*), ZFIN (zebrafish *gatad1*), Alliance of Genome Resources.

Summary of Supported vs. Refuted / Open Hypotheses

- **Supported:** CMD2B = autosomal-recessive DCM caused by biallelic GATAD1 (p.Ser102Pro); GATAD1 is a nuclear H3K4me3-reader/chromatin-complex subunit; the mutation disrupts a Ser102 phospho-14-3-3 motif and GATAD1 nuclear localization; GATAD1 regulates cardiac substrate metabolism; zebrafish recapitulate DCM.
- **Refuted / qualified:** "simple cardiomyocyte loss-of-function causes the disease" — **refuted** in mouse cKO
([P 39641830](#)); mechanism is more likely aberrant-function/mislocalization.
- **Resolved this review:** founder allele nomenclature and status — GATAD1 NM_021167.5:c.304T>C (p.Ser102Pro), ClinVar VCV31656/rs387907188, **Pathogenic, absent from gnomAD**.
- **Open / data-unavailable:** population carrier frequency; CMD2B-specific penetrance %, survival, QoL; modifier genes; existence of additional families beyond the index kindred; iPSC and knock-in (p.Ser102Pro) models.

Limitations

Human evidence rests on one consanguineous kindred; mechanism is pieced together from in-vitro and cross-species models with a notable mouse/zebrafish discordance. Epidemiologic, prognostic, and treatment-response data specific to CMD2B do not exist and were substituted with clearly-labeled DCM-class information.

Key References (PMIDs)

- 21965549 — Theis et al. 2011: discovery of GATAD1 in AR-DCM [human clinical].
- 38605029 — Rustemi et al. 2024: Ser102/14-3-3, nucleocytoplasmic transport [in vitro].
- 26841866 — Varier et al. 2016: EMSY/KDM5A/SIN3B complex, GATAD1 subunit [in vitro].
- 28955713 — Yang et al. 2016: adult zebrafish *gatad1* DCM model [model organism].

- 39641830 — Pang et al. 2024: cardiomyocyte Gatad1 cKO — no cardiomyopathy [model organism].
- 39626862 — Cai et al. 2025: GATAD1 controls FAO/glucose oxidation, I/R injury [model organism].
- 38664609 / 40200748 / 39833651 / 39855353 — DCM genetic testing & family screening context.
- 31431100 — DCM natural history by EF stratum [human clinical, DCM-class].

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